General instructions for all candidates

- Complete the core requirements first. Extra features are optional.
- Preferred language: Python
- Use Pandas, NumPy, scikit-learn, and other relevant libraries.
- Submission requirements
- Source code, README with execution instructions, sample output, and results from the
- provided test cases.
- Important: Candidates may use synthetic data if no dataset is supplied. For tasks involving machine learning, the expected predictions are based on the trained model and test data; they do not need to match one universal numerical answer. The test cases below specify the behaviour that must be demonstrated.

Problem: Customer Segmentation Using K-Means

Unsupervised Learning · Clustering

Problem statement

Develop a customer segmentation system that groups customers based on purchasing behaviour. The business wants to identify groups such as low spenders, frequent shoppers, and high-value customers to support targeted marketing.

Requirements

1. Load a CSV file containing customer_id, annual_income, spending_score, and optionally purchase_frequency.
2. Handle missing values and scale numerical features where appropriate.
3. Allow the user to specify the number of clusters, \(K\).
4. Train a K-Means clustering model.
5. Assign a cluster ID to each customer.
6. Display the number of customers in each cluster.
7. Visualize clusters in a two-dimensional scatter plot.
8. Explain the characteristics of each cluster using feature averages or other summary statistics.

Test cases

Test case	Input	Expected behaviour
TC1	Dataset with clearly separated low-, medium-, and high-spending groups	Produce clusters that broadly reflect the groups
TC2	\(K=3\)	Assign each valid customer to one of three cluster IDs
TC3	\(K=1\)	Produce one cluster if supported by the implementation
TC4	\(K=0\) or negative	Reject input
TC5	Missing income for one customer	Apply a documented strategy or reject the affected record
TC6	Dataset containing fewer valid records than requested clusters	Handle the invalid configuration gracefully

Acceptance criteria: Every valid customer receives a cluster assignment, cluster summaries are produced, and invalid configurations are handled correctly.